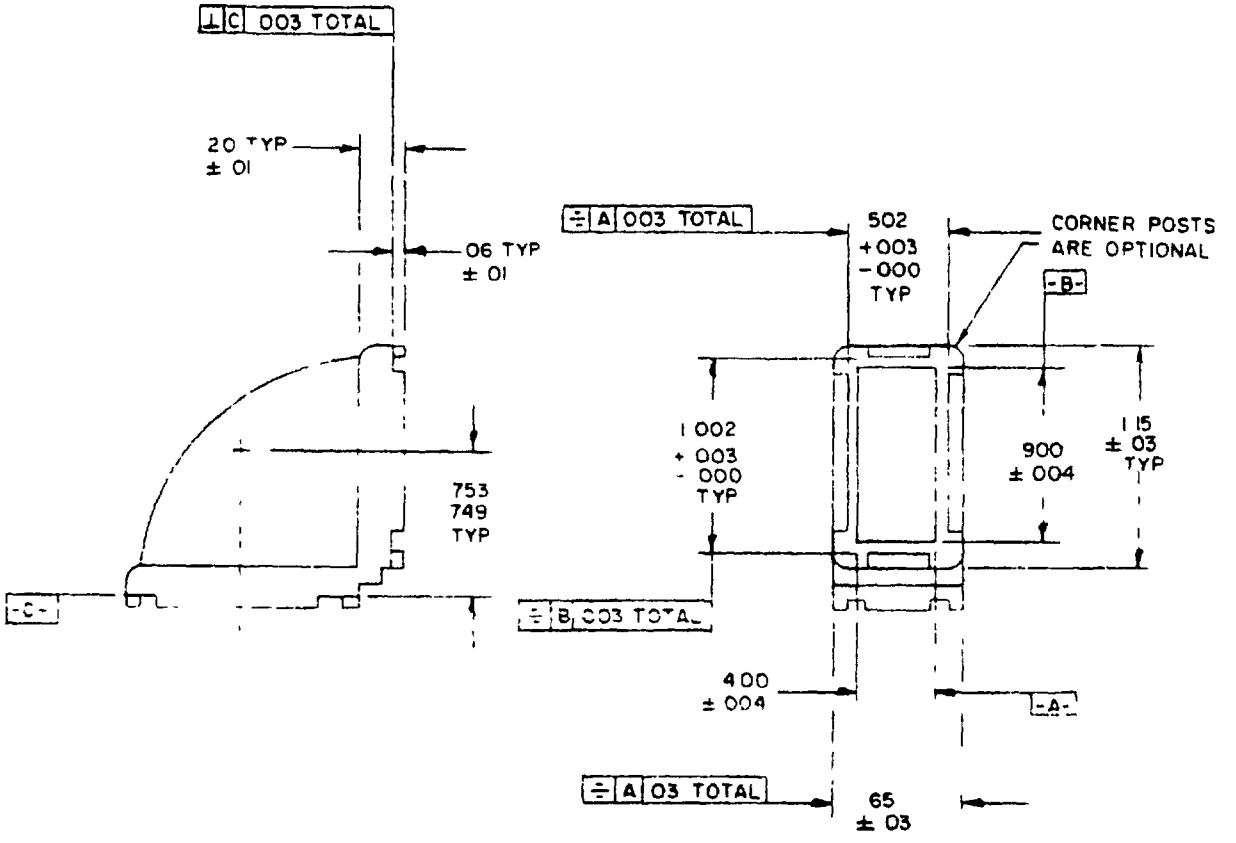
MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLY, RIGID, CAST, 90-DEGREE H-PLANE BEND, FULL CORRAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide assembly described herein shall consist of this document and the latest issue of Specification MIL-W-3970.



INCHES	MM	INCHES	MM	INCHES	MM
003	.08	06	1 5	743	18 87
.004	10	20	5.1	753	19 13
.01	.3	400	10 16	.900	22.86
.03	8	502	12.75	1 002	25 45
		65	16.5	1.15	29.2

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm

FIGURE 1. Dimensions for full corral configuration.

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-51 [illegible]

Physical characteristics:
Dimensions and configuration: See figure 1.
Operating temperature range: -55° to +200°C.
Material: 6712.0, temper F, aluminum per QQ-A-601 or 6712 per QQ-A-371.
Surfaces (sandblasting after machining permissible):
Internal: 63 $\sqrt{}$ microinches per ANSI B46.1.
External: 125 $\sqrt{}$ microinches per ANSI B46.1.

Fungus resistance: Material is nutrient to fungus shall not be used.

Pressurization: 45 lbf/in^2 gage maximum.

Penetrant inspection: No indication of any cracks or any location when tested per MIL-I-6866, type I, method b

Electrical characteristics:
Frequency range: 8.2 to 12.4 GHz.
VSWR: 1.05:1 maximum.
Power:
Peak - 250 kW.
Average - 10 kW.

Part number: M3970/28-001.

NOTE: The mating waveguides are M85/1-077, -078, and -178.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI,
Navy - EC, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT
Navy - MC, OS, AS
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5985-0881-8)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE
MIL-W- 3770/2F

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

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